Bayesian estimation of dissolved-oxygen reduction in Heretaunga groundwater

A two-endmember mixing and dual-pathway kinetic model: workflow, validation, posterior inference, and limitations

Technical report and manuscript working draft

Analysis version: regional 39-well DO likelihood, November 2026
Status: final model, grouped cross-validation and seven-fit sensitivity analysis complete; three selected DO–MRT figures (script plots 1, 2 and 4), plus a separate conceptual source-decay illustration; source-specific flowpath ages unresolved
Source code: Oxygen-Reduction-Joint-Model (script 05 exports five DO–MRT figures; all five appear in this report)
Data: local, non-public prepared groundwater dataset (not included)

Main result. Our selected scientific model combines first-order aquifer oxygen consumption with *uncertain* river-entry DO, using a truncated Normal prior centred at $7.5\,\mathrm{mg\,L^{-1}}$ (SD $1.5\,\mathrm{mg\,L^{-1}}$). Reanalysis of the saved posterior fit in the completed sensitivity workflow gives $k_1 = 0.0639\,\mathrm{yr^{-1}}$ (95% credible interval 0.0458–0.0857). The inferred land-surface recharge (LSR) water-table DO is $3.26\,\mathrm{mg\,L^{-1}}$ (1.31–5.57). The first-order rate changes little under tested LSR saturation/prior assumptions, but more with assumed uncertainty in groundwater mean residence time (MRT). Source-specific groundwater ages remain the major structural uncertainty.

Prepared for collaborator correspondence, internal technical review, and adaptation into a scientific manuscript. Numerical results derive from the final project outputs discussed in November 2026; some diagnostic plots require the user's local saved files.

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1 Executive summary

The aim was to identify oxygen-reduction processes in the Heretaunga groundwater system while distinguishing two potential components of dissolved-oxygen (DO) depletion: (i) reduced oxygen availability in land-surface recharge (LSR) by the time it reaches the water table, and (ii) DO consumption during subsequent aquifer residence. The analysis couples a two-endmember mixing model, informed by $\delta^{18}O$ and chloride, with alternative zero- and first-order aquifer DO kinetics and uncertain groundwater mean residence times (MRTs).

The final regional calibration contains **39 groundwater DO observations**. River-proximal site `2801` and the shallow, water-race-adjacent site `2130` are excluded from the DO likelihood, but both retain their conservative tracer observations. A separate southern outlier, `16758`, is removed from the prepared dataset. Four full-data model variants were fitted (two river DO assumptions by two kinetic orders), and their predictive performance was compared using five-fold grouped held-out DO cross-validation (20 additional model fits).

The final scientific interpretation adopts **first-order aquifer kinetics and an uncertain river-entry DO boundary**, with $C_R \sim \mathcal{N}(7.5, 1.5^2)\,\mathrm{mg\,L^{-1}}$ truncated to 0–14.65. Although the fixed-river first-order alternative had a small advantage in grouped CV RMSE (median $0.0342\,\mathrm{mg\,L^{-1}}$), treating $7.5\,\mathrm{mg\,L^{-1}}$ as an exactly known entry condition is not physically compelling. The earlier full-data summary CSV reported $k_1 = 0.0632\,\mathrm{yr^{-1}}$; the later sensitivity-run baseline (now script 06), extracted from the saved preferred CmdStanR fit, gave $k_1 = 0.0639\,\mathrm{yr^{-1}}$ (95% CrI 0.0458–0.0857). This minor discrepancy is documented explicitly below, rather than silently combining summaries. All original full-data and sensitivity fits passed the reported MCMC checks.

Seven additional one-factor-at-a-time full-data fits examined recharge temperature (10 and 20°C), alternative LSR survival priors, and MRT log-SD (0.10 and 0.50; baseline 0.25). Estimated rate changes were under 0.2% for temperature, −1.0 to +3.4% for LSR prior choices, and −4.6 to +18.5% for MRT uncertainty. The corresponding land-surface-to-water-table DO deficit is more sensitive to the assumed saturation concentration than the inferred water-table concentration.

The most important unresolved question is structural: the model estimates a separate *final mixing fraction* at each well but assigns the *same well-specific MRT* to river and LSR components. It therefore does not simulate distributed LSR inflow along a flowpath or the shorter residence time of recharge entering downstream. This may affect partitioning of the observed oxygen depletion between LSR entry conditions and aquifer reaction, and is a priority for scientific interpretation rather than further MCMC chain length.

Five script-generated DO–MRT plots (Figures 5–9) distinguish the common first-order decay relationship after adjustment for recharge composition, observed versus fitted groundwater DO, raw observations against a pure-river reference, the river-attributable DO component, and the additive river/LSR contributions. The uncertainty intervals represent different quantities and must not be treated as interchangeable.

2 Research question and analysis strategy

2.1 Objectives

1. Quantify the contribution of river recharge and land-surface recharge to sampled groundwater using conservative tracers.
2. Estimate a common aquifer DO consumption rate under zero- and first-order reaction assumptions.
3. Infer the DO of LSR reaching the water table, and the associated oxygen deficit relative to

assumed DO saturation at the land surface.

4. Evaluate sensitivity to the assumed river-entry DO boundary and compare predictive skill using held-out groundwater DO measurements.
5. Test LSR saturation, oxygen-survival prior and groundwater MRT uncertainty, while distinguishing these tests from unresolved source-specific age structure.
6. Identify which estimates are supported by the observations and which depend on untested hydrogeological structure.

2.2 Evolution of the workflow

The modelling programme progressed from more elaborate mixing and reaction formulations toward a parsimonious two-source framework. Earlier development considered additional mixing tracers, including sodium and excess air, paired DO–excess nitrogen formulations, and denitrification-related alternatives. Including sodium caused MCMC convergence problems, whereas excess air provided limited added value and generated conflicting results; neither is retained in the adopted tracer set. Tests of spatially varying LSR oxygen availability did not materially improve the earlier grouped CV scores (reported RMSE approximately 1.815 versus 1.812 $\mathrm{mg\,L^{-1}}$ for common and variable LSR DO, respectively). These were *developmental comparisons*, not substitutes for the final regional 39-well validation.

The final implementation retains δ^{18}O and chloride only as conservative mixing tracers, a single inferred LSR DO at the water table, and one common aquifer reaction law applied independently to each source before their contributions are mixed. This minimises unnecessary degrees of freedom while preserving uncertainty in endmember composition, source fractions, groundwater ages and rates. The Positron revisions reorganised the R scripts. The report includes all five DO–MRT figures from script 05, directly from the selected first-order posterior; none is a separately fitted regression or a copied screenshot. This PDF is a snapshot: to update it after regenerating plots, recompile the LaTeX source. The conceptual source-decay figure in this report is a separate explanation. The core Stan likelihood and fitted kinetic model are unchanged.

3 Observations, screening and inclusion rules

The private input is `data/Uwe_HP_data_prepared.csv`. The fitting scripts require well identifier, depth, DO, reported MRT, δ^{18}O and chloride; they retain records with finite non-negative DO and finite non-negative reported MRT. The initial data preparation removes site `16758` as the identified far-south anomalous well.

After basic screening, 41 groundwater records remain, but only **39 DO responses** contribute to the final primary model:

- `2801`: measured groundwater DO 10.2 $\mathrm{mg\,L^{-1}}$, assumed MRT zero. It is river-proximal but is *not* treated as a direct measurement of river-entry DO. Its actual measured DO is preserved and masked in the model likelihood.
- `2130`: measured DO 8.8 $\mathrm{mg\,L^{-1}}$, reported MRT approximately 8 years. This shallow well is adjacent to a water race in the western system and may not represent regional groundwater. Its DO is excluded on a hydrogeological representativeness basis, not replaced with a modelled value.
- At both sites, isotope and chloride data are retained for source mixing; the masking acts on the DO likelihood, not on the groundwater record as a whole.

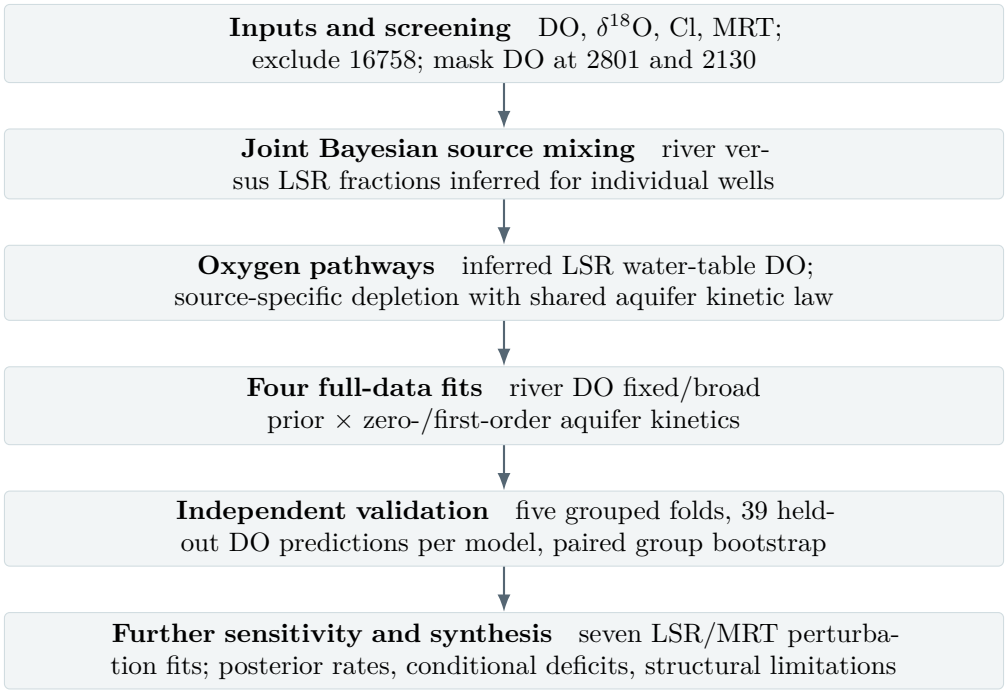


Figure 1: Final analytical workflow. Conservative tracers remain available even when a well's DO is withheld from calibration. This diagram describes the implemented workflow, not a causal flowpath simulation.

The choice to exclude `2130` follows investigation of the systematic underprediction of this high-DO observation. It should be documented transparently when reporting regional inference. The previous 40-well analysis (which includes 2130 DO) remains historical sensitivity evidence, not the preferred regional calibration.

4 Mathematical model

4.1 Conservative two-endmember mixing

For each well i, the inferred land-surface recharge fraction f_i is restricted to $[0, 1]$, with river recharge fraction $1 - f_i$. Expected conservative tracer concentrations are

$$\widehat{\delta^{18}O}_i = (1 - f_i)\,\delta^{18}O_R + f_i\,\delta^{18}O_{\mathrm{LSR}}, \tag{1}$$

$$\widehat{\mathrm{Cl}}_i = (1 - f_i)\,\mathrm{Cl}_R + f_i\,\mathrm{Cl}_{\mathrm{LSR}}. \tag{2}$$

Both source endmembers are estimated with informative, bounded prior distributions. The priors in the current Stan formulation have centres -7.79 and -6.58 per mil for river and LSR δ^{18}O, and 4.9 and 60 mg L^{-1} for river and LSR chloride, respectively; corresponding prior scales are 0.24, 0.40, 3 and 15. The Stan code also constrains source ranges. Each well has an independent $f_i \sim \mathrm{Beta}(1, 1)$ prior, but f_i is estimated *jointly* from its tracers and DO, not solely from conservative tracers. Isotope uncertainty includes an inferred LSR-fraction-dependent contribution, and chloride uncertainty includes a measurement-error component plus inferred residual dispersion.

An important distinction: the model permits higher f_i at greater distance from a river if the observations support it, but contains **no explicit distance-to-river covariate, monotonic mixing constraint or transport equation**. The mixing fractions describe the sampled composition of each well, not the chronology of recharge addition.

4.2 Oxygen boundary conditions

The assumed DO of rainfall/LSR at the land surface is freshwater saturation at 15°C, approximately $C_{\text{sat}} \simeq 10\,\text{mg L}^{-1}$, assuming atmospheric pressure ratio one. A common, unknown survival fraction s determines the DO concentration where LSR enters the groundwater system:

$$C_{\text{LSR},WT} = sC_{\text{sat}}, \qquad s \sim \text{Beta}(1,1), \qquad \Delta C_{\text{LSR}} = C_{\text{sat}} - C_{\text{LSR},WT}. \tag{3}$$

The inferred ΔC_{LSR} is termed the *LSR oxygen deficit*. Without independent vadose residence-time information, it is *not* a separate fitted vadose-zone reaction rate.

Two river-entry boundary treatments are evaluated:

$$\text{Scenario A:} \quad C_R = 7.5\,\text{mg L}^{-1} \quad \text{(fixed)}, \tag{4}$$

$$\text{Scenario C:} \quad C_R \sim \mathcal{N}(7.5, 1.5^2)\,\text{mg L}^{-1}, \quad 0 \le C_R \le 14.65\,\text{mg L}^{-1}. \tag{5}$$

The constrained upper bound allows the possibility of colder, more oxygenated river water. River-entry DO is a global boundary parameter; the model does not use well 2801's measured groundwater DO as that boundary condition.

4.3 Aquifer oxygen consumption

For aquifer exposure duration t, the competing reaction laws are fitted *separately*:

$$\text{Zero order:} \quad C^{(0)}(t; C_0) = \max(0, C_0 - k_0 t), \qquad [k_0] = \text{mg L}^{-1}\,\text{yr}^{-1}, \tag{6}$$

$$\text{First order:} \quad C^{(1)}(t; C_0) = C_0 \exp(-k_1 t), \qquad [k_1] = \text{yr}^{-1}. \tag{7}$$

In the code, $k_0 = k_{\text{ref}} \times 7.5\,\text{mg L}^{-1}$ and $k_1 = k_{\text{ref}}$; both models use the same positive reference-rate prior specification but have different physical units and kinetics. There is *one* aquifer consumption rate in each model, shared between the two recharge components.

For the selected first-order form, predicted mean DO at well i is

$$\widehat{\text{DO}}_i = (1 - f_i) C_R \exp(-k_1 t_i) + f_i C_{\text{LSR},WT} \exp(-k_1 t_i) \tag{8}$$

$$= \left[(1 - f_i) C_R + f_i C_{\text{LSR},WT}\right] \exp(-k_1 t_i). \tag{9}$$

For zero order, each source concentration is reduced with its own $\max(0, \cdot)$ before the depleted contributions are mixed. Thus, mixing and zero-order reduction must not be interchanged algebraically without retaining the non-negativity operation.

The observed DO likelihood is Student-t with four degrees of freedom and scale $\sqrt{0.2^2 + \sigma_{\text{DO}}^2}\,\text{mg L}^{-1}$. This allows greater robustness to outliers than a normal residual model. DO is not a conservative mixing tracer.

4.4 Groundwater residence-time treatment

Positive reported MRTs are treated as uncertain rather than exact:

$$t_i = \exp\left[\log(\text{MRT}_i) - \frac{\sigma_t^2}{2} + \sigma_t z_i\right], \qquad z_i \sim \mathcal{N}(0,1), \quad \sigma_t = 0.25. \tag{10}$$

The mean-preserving lognormal parameterisation centres the prior arithmetic mean of t_i on the reported MRT. Site 2801, assigned MRT zero, has $t = 0$. Most importantly, the model assumes **the same effective aquifer residence time for both recharge sources at a given well**. This simplifying assumption is revisited in Section 9.

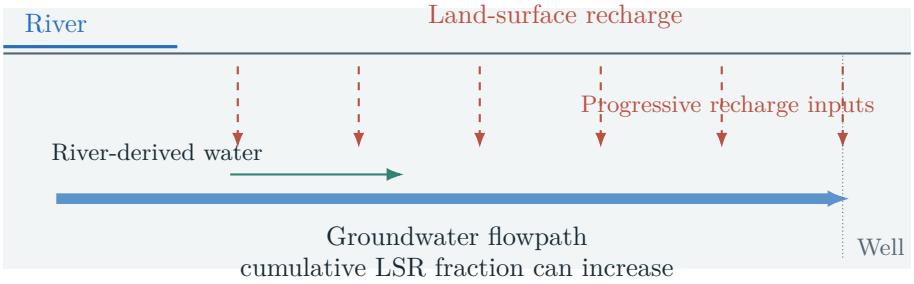


Figure 2: Conceptual groundwater flowpath. Increasing LSR contribution with distance can be represented by increasing well-specific f_i. The time at which individual LSR increments entered the aquifer, however, is not represented by the common well-specific MRT. The drawing is conceptual, not a cross-section inferred from mapped data.

5 Bayesian computation and diagnostics

The model was implemented in Stan via `stan/04_river_DO_prior_sensitivity.stan`, called by R/CmdStanR. Four independent full-data posterior fits compare two river boundary assumptions with zero- and first-order kinetics. Each full-data model uses four chains, 1,400 warm-up and 1,400 retained sampling iterations per chain: 5,600 retained posterior draws per model. The configuration uses target acceptance `adapt_delta` = 0.98 and maximum tree depth 13.

Convergence was assessed with rank-based $\widehat{R}$, bulk effective sample size (ESS), divergent transitions, maximum tree-depth hits and energy-based fraction of missing information (E-BFMI). All four final full-data models passed the reported gates. Table 1 reproduces the final outputs.

Table 1: Reported full-data Stan diagnostics for the final four variants. The ESS values are the minima across parameters monitored by script 04; they are not limited to the main scientific parameters.

River boundary	Aquifer order	Max $\widehat{R}$	Min bulk ESS	Divergences	Depth hits	Min E-BFMI
Fixed 7.5	Zero	1.01	713	0	0	0.445
Fixed 7.5	First	1.01	1,031	0	0	0.478
Broad prior	Zero	1.00	852	0	0	0.444
Broad prior	First	1.00	874	0	0	0.479

For the fixed-boundary first-order *comparator*, the key parameters had even larger bulk/tail ESS: 4,265/4,373 for the aquifer rate, 3,628/3,013 for LSR DO deficit, and 5,070/4,304 for residual DO dispersion. The associated maximum $\widehat{R}$ was 1.00 to the printed precision. The resulting numerical Monte Carlo uncertainty is much smaller than parameter uncertainty; longer chains are not indicated by the available diagnostics. Convergence does not, by itself, establish the correctness of the hydrogeological model.

6 Grouped held-out cross-validation

6.1 Design

Cross-validation used five matched folds, retaining all observations from a named well family in the same fold where applicable. The 39 eligible DO observations are scored once per model; the four variants use identical fold assignments. For a held-out well, its DO likelihood indicator is switched off, but its tracer measurements and reported MRT remain available. Thus, the predictive assessment is *held-out DO conditional on available chemistry and age information*, not prediction at a wholly unsampled location.

Four model variants across five folds result in 20 CV fits. Two fold-3 fits initially had insufficient effective sample size and were selectively resampled for more iterations; the completed set passed the CV diagnostics and supported grouped bootstrap comparison. A paired bootstrap resamples the prespecified well groups, not independent posterior observations, and compares RMSE or MAE for the same held-out well set. The reported 95% bootstrap intervals quantify variability of the paired prediction metrics under this grouping and are not posterior credible intervals for a kinetic coefficient.

6.2 Paired results

Let $\Delta\text{RMSE} = \text{RMSE(candidate)} - \text{RMSE(comparator)}$. Negative values favour the candidate. Table 2 and Figure 3 summarise the completed paired bootstrap results.

Table 2: Paired grouped-bootstrap predictive differences (mg L^{-1}); intervals are 2.5th–97.5th percentiles. Negative candidate-minus-comparator values favour the candidate.

Candidate minus comparator	Median ΔRMSE [95% interval]	Median ΔMAE [95% interval]
Broad minus fixed (first order)	0.0342 [0.00564, 0.0751]	0.0204 [-0.00873, 0.0664]
Broad minus fixed (zero order)	-0.0482 [-0.132, 0.0480]	-0.0555 [-0.145, 0.0323]
First minus zero (broad river prior)	-0.0473 [-0.133, 0.0321]	0.0192 [-0.0553, 0.0818]
First minus zero (fixed river DO)	**-0.132 [-0.238, -0.0126]**	-0.0593 [-0.162, 0.0265]

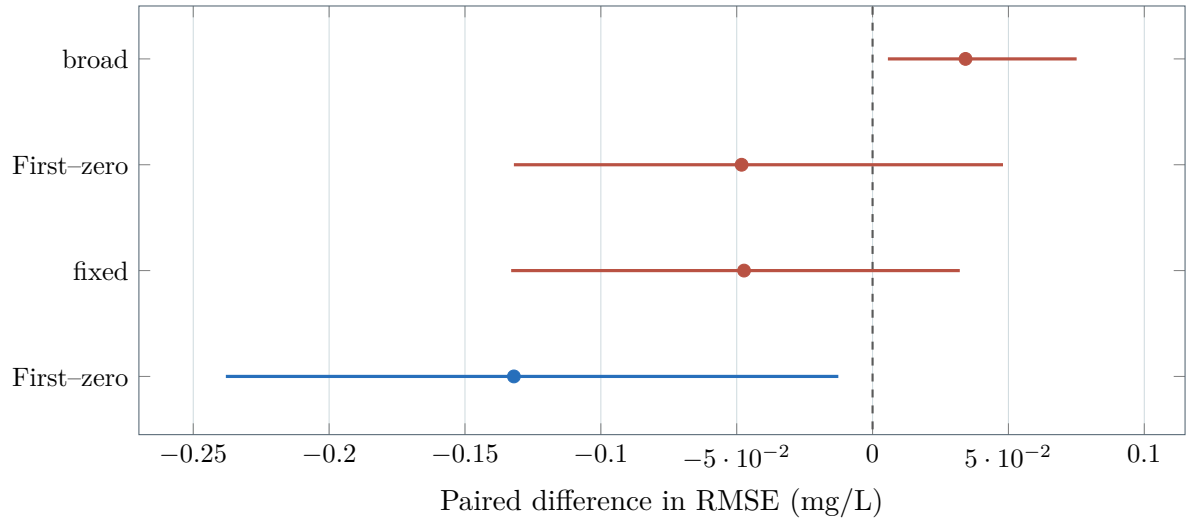


Figure 3: Median differences in held-out RMSE and grouped-bootstrap 95% intervals. A negative difference indicates lower prediction error for the candidate model. The fixed-river, first-order formulation performs better than fixed-river zero-order by RMSE under this validation design, while other differences are generally small and/or less conclusive.

The fixed-river first-order model improves RMSE relative to fixed-river zero-order by a median $0.132\,\text{mg L}^{-1}$, with a paired 95% interval that excludes zero. The corresponding MAE interval includes zero; the result should therefore not be presented as a large or unequivocal performance gain. Under first-order kinetics, widening the river DO boundary to the broad prior slightly *worsens* RMSE (median $+0.0342\,\text{mg L}^{-1}$). For *predictive RMSE alone*, the fixed-river first-order option is marginally better. For *scientific interpretation*, we select the uncertain-river first-order option because a fixed $7.5\,\text{mg L}^{-1}$ entry boundary would treat a variable physical quantity as perfectly known. In the broad-river comparison, first-minus-zero order improves median RMSE by $0.0473\,\text{mg L}^{-1}$, but the paired interval includes zero; hence preference for first-order kinetics is not statistically decisive under every boundary assumption. Absolute 39-well CV scores should be read

from the corresponding final `cv_scores.csv` before submitting a paper; they are not reproduced here because that final file was not supplied for this synthesis.

7 Final posterior parameter estimates

Table 3 shows the fitted full-data posterior medians and central 95% credible intervals for all four variants. Rate units differ between kinetic orders and should not be compared as though they were the same physical quantity.

Table 3: Earlier full-data posterior CSV summaries (before the later sensitivity-run saved-fit re-summarisation), medians with 95% credible intervals. River and LSR DO units are $\mathrm{mg\,L^{-1}}$; zero-order rates are $\mathrm{mg\,L^{-1}\,yr^{-1}}$ and first-order rates are $\mathrm{yr^{-1}}$. The broad-river first-order fit is the scientific primary.

Boundary	Order	River-entry DO	Aquifer rate	LSR DO at WT	LSR DO deficit
Fixed	Zero	7.50 (fixed)	0.235 [0.189, 0.293]	3.62 [0.462, 5.45]	6.38 [4.55, 9.54]
Fixed	First	7.50 (fixed)	0.0625 [0.0484, 0.0812]	3.28 [1.42, 5.47]	6.72 [4.53, 8.58]
Broad	Zero	6.35 [5.02, 8.47]	0.188 [0.125, 0.294]	3.98 [0.825, 5.43]	6.02 [4.57, 9.18]
Broad	**First**	**7.66 [5.75, 9.74]**	**0.0632 [0.0453, 0.0851]**	**3.21 [1.25, 5.55]**	**6.79 [4.45, 8.75]**

7.1 Selected first-order solution and posterior consistency

The scientific primary has an uncertain river-entry DO boundary. The *later baseline* from the sensitivity workflow (current script 06), which reads the preferred posterior fit originally saved by current script 02, gives

$$\boxed{k_1 = 0.0639\,\mathrm{yr^{-1}} \quad (95\%\ \mathrm{CrI}:\ 0.0458\text{–}0.0857\,\mathrm{yr^{-1}}).} \tag{11}$$

Its corresponding LSR water-table DO is $3.26\,\mathrm{mg\,L^{-1}}$ (95% CrI 1.31–5.57) and its LSR DO deficit is $6.78\,\mathrm{mg\,L^{-1}}$ (4.46–8.72). The effective first-order half-life at the posterior median is $\ln(2)/k_1 \simeq 10.8$ years; this is a kinetic interpretation, not a measured groundwater transit time.

Consistency note. The earlier full-data summary CSV (Table 3) reported $k_1 = 0.0632$ (0.0453–0.0851), LSR water-table DO 3.21 (1.25–5.55) and deficit 6.79 (4.45–8.75). The sensitivity-run baseline reported 0.0639, 3.26 and 6.78 from the saved fit. These differences are small but need provenance reconciliation before formal publication. The renumbered script 04 reads the preferred RDS directly and warns when its rate disagrees with the earlier summary CSV; the new script 05 independently prints a four-parameter summary from the same saved RDS. The previously reported broad-boundary river DO posterior is 7.66 (5.75–9.74) $\mathrm{mg\,L^{-1}}$, which has not been independently re-summarised for the present report.

The LSR deficit is the *difference between assumed surface DO and inferred water-table DO*. It is a model-conditional concentration deficit, not a measured vadose-zone oxygen consumption rate or proof of where the oxygen was removed.

7.2 Five DO–MRT plots: observed concentrations, fitted kinetics and recharge contributions

The active `scripts/05_DO_kinetic_trajectoriesandsummary.R` exports five PNGs under `plots/final_39well/`. All five appear below in the same sequence as the script. They are complementary views of one jointly fitted first-order Bayesian model, not five independent regressions.

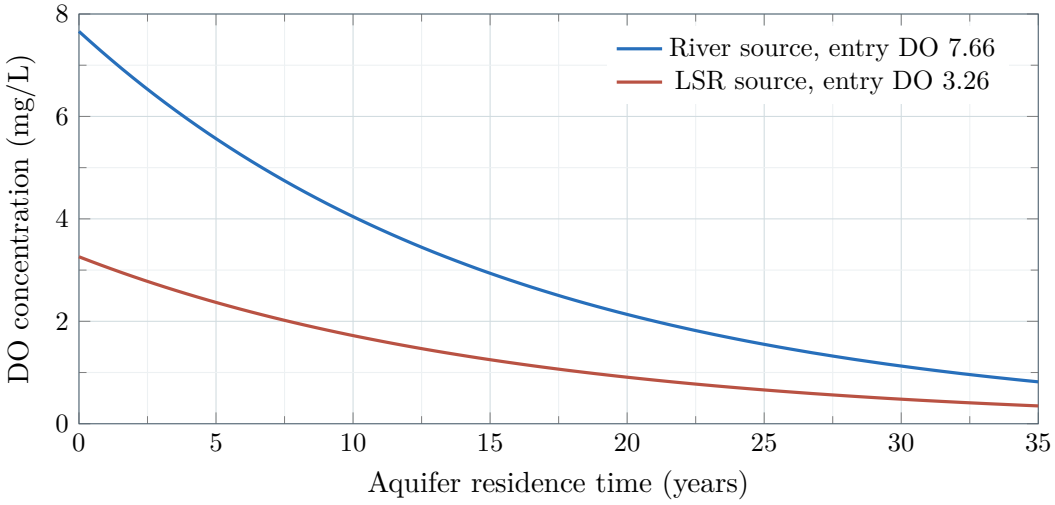


Figure 4: Illustrative first-order DO decay using the *later reported* median rate $0.0639\,\mathrm{yr}^{-1}$ and input DO medians for the uncertain river-boundary model. The curves show model behaviour for isolated source components; they are not observed concentration transects, not posterior credible bands, and do not represent progressively added LSR.

1. **Plot 1: mixing-adjusted measured DO and fitted first-order survival (Figure 5).** The initial oxygen concentration for well i is inferred from its river/LSR mixture:

$$C_{0,i} = (1 - f_i)C_{R,\mathrm{entry}} + f_i C_{\mathrm{LSR,WT}}, \tag{12}$$

$$\mu_i / C_{0,i} = \exp(-k_1 t_i). \tag{13}$$

   Here f_i is its LSR fraction, and t_i is the modelled effective aquifer residence time. Each coloured marker is the measured DO divided by the posterior-inferred $C_{0,i}$ (median across posterior draws). The point whiskers are 95% credible intervals *from the inferred denominator only*; they are not intervals for the measurement error or full posterior predictive distribution. The dashed curve and ribbon are the posterior-median common survival $\exp(-k_1 t)$ and 95% credible interval in the fitted rate. The curve was *not fitted independently to these adjusted points*. The horizontal axis shows reported MRT; the model instead uses latent uncertain MRT. The adjusted data are model-conditioned, not independent validation of k_1, and can exceed unit survival when observed DO is larger than the inferred entry concentration.

2. **Plot 2: measured versus fitted mixed DO (Figure 6).** Open circles are measured DO included in the likelihood; crosses are measurements excluded from the DO likelihood at `2801` (10.2 mg/L; nominal MRT 0) and `2130` (8.8 mg/L). Coloured points are posterior medians of sample-specific fitted mixed DO, with colour representing LSR fraction. Whiskers are 95% *posterior predictive* intervals based on replicated sample DO and include observation-level variability. This is the principal in-sample goodness-of-fit diagnostic.

3. **Plot 3: raw groundwater DO against the pure-river reference (Figure 7).** Markers show raw measured mixed-sample DO; the dashed line and blue ribbon show the posterior median and 95% credible interval of the 100%-river first-order reference $C_{R,\mathrm{entry}} \exp(-k_1 t)$. The river curve follows the fitted parameters but is *not* itself a fitted regression through the observations or a predictive interval for mixed samples.

4. **Plot 4: inferred river-attributable DO in mixed samples (Figure 8).** Markers represent the river-derived contribution $(1 - f_i)C_{R,\mathrm{entry}} \exp(-k_1 t_i)$, derived from joint posterior draws; whiskers are credible intervals for this contribution. The reference curve is for unweighted, 100%-river-derived groundwater. For large f_i, contributions necessarily lie below the reference. The gap does not independently estimate reaction loss or the quality of model fit.

5. **Plot 5: additive river and LSR contributions (Figure 9).** Source contributions are

$$C_{R,i} = (1 - f_i)C_{R,\text{entry}} \exp(-k_1 t_i), \tag{14}$$

$$C_{\text{LSR},i} = f_i C_{\text{LSR,WT}} \exp(-k_1 t_i), \tag{15}$$

$$\mu_i = C_{R,i} + C_{\text{LSR},i}. \tag{16}$$

   The blue river and red LSR segment heights are *posterior means*, so they add exactly to the black posterior-mean fitted total DO. Open circles and crosses show observations, while grey whiskers are 95% *credible intervals for fitted mean DO*, not posterior predictive ranges. This plot explains the inferred composition of each fitted oxygen concentration.

Together, Plot 1 shows the first-order rate relationship after accounting for estimated source composition; Plot 2 assesses the fit to raw mixed-groundwater DO; Plot 3 shows a hypothetical river-only kinetic reference; Plot 4 isolates the river portion of fitted DO; and Plot 5 illustrates how both source components sum to the fitted total. None is independent validation of the reaction rate. Grouped held-out cross-validation (Section 6) addresses predictive performance more directly. The common-MRT assumption for both recharge components remains a structural limitation.

Figure provenance. The five panels embed the current script-generated `01_DO_` through `05_DO_` PNGs, never phone-uploaded screenshots. GitHub does not include these gitignored local outputs; a compiled preview without them displays explicit missing-output notices. Regenerate the figures locally, then recompile this report.

Script-generated figure unavailable in this build

This is not a replacement image.

Run the active Positron plot script from the project root, then recompile the LaTeX report to embed the generated output.

Expected output folder:
`plots/final_39well/`

Figure 5: Plot 1: mixing-adjusted measured DO and the common first-order survival curve. Points are measured DO divided by each well's posterior-inferred entry DO for its source mixture. Whiskers show 95% credible intervals in this normalisation only. The dashed $\exp(-k_1 t)$ curve and blue band use the common first-order rate estimated in the full joint model, not a separate regression through these model-conditioned points. The x-axis uses reported MRT, whereas the likelihood uses uncertain true MRT.

Script-generated figure unavailable in this build

This is not a replacement image.

Run the active Positron plot script from the project root, then recompile the LaTeX report to embed the generated output.

Expected output folder:
`plots/final_39well/`

Figure 6: Plot 2: observed and fitted groundwater DO. Open circles and crosses show observed mixed-sample DO, with crosses excluded from the DO likelihood at sites `2801` and `2130`. Coloured markers are the fitted mixed-DO posterior medians, coloured by inferred LSR fraction; the whiskers are 95% posterior predictive intervals including observation residual variability.

Script-generated figure unavailable in this build

This is not a replacement image.

Run the active Positron plot script from the project root, then recompile the LaTeX report to embed the generated output.

Expected output folder:
`plots/final_39well/`

Figure 7: Plot 3: measured mixed DO versus the 100%-river-only first-order reference. Coloured markers are raw groundwater DO. The dashed curve and blue band show the model-derived pure-river trajectory and its 95% credible interval, *not* the expected mixed-sample DO or its prediction interval.

Script-generated figure unavailable in this build

This is not a replacement image.

Run the active Positron plot script from the project root, then recompile the LaTeX report to embed the generated output.

Expected output folder:
`plots/final_39well/`

Figure 8: Plot 4: inferred river-attributable DO within mixed groundwater. Markers and 95% credible whiskers represent the joint-posterior river contribution, $(1 - f_i)C_{R,i}$, where $C_{R,i}$ is the remaining river-endmember DO after aquifer reduction. The pure-river reference is unweighted: lower marker values follow from the inferred LSR fraction and are not independently measured river DO.

Script-generated figure unavailable in this build

This is not a replacement image.

Run the active Positron plot script from the project root, then recompile the LaTeX report to embed the generated output.

Expected output folder:
`plots/final_39well/`

Figure 9: Plot 5: fitted river and LSR DO contributions. Blue and red stacked segments are posterior-mean contributions from river and LSR recharge, and their exact sum gives the black fitted-total point. Open circles and crosses are measured groundwater DO; grey whiskers give 95% credible intervals for the fitted mean rather than full predictive intervals.

The separate conceptual first-order source-decay illustration (Figure 4) is not one of these five script-generated DO–MRT figures. The previously reported 0.0632 versus saved-fit 0.0639 yr^{-1} rate-summary discrepancy still needs reconciliation before formal publication.

7.3 Boundary sensitivity and posterior dependence

The first-order rate changes little between the fixed boundary and the initially reported uncertain-river fit (0.0625 versus 0.0632 yr^{-1}); the later saved-fit baseline was 0.0639 yr^{-1}. LSR water-table DO and deficit also remain similar at the posterior median. This is evidence of stability *within the tested family of models*, not proof of independence from all river-water oxygen dynamics.

The fixed-river first-order comparator has a posterior correlation of approximately −0.397 between LSR oxygen deficit and aquifer rate, while the selected broad-river first-order formulation has a smaller negative correlation (approximately −0.192). The sign is physically coherent: greater DO loss before aquifer entry can, to some extent, be offset by slower aquifer reduction. In the broad-river sensitivity, river DO and aquifer rate are correlated positively (+0.838 for zero-order; +0.544 for first-order), indicating appreciable parameter compensation when the input DO is not fixed. The smaller first-order correlation is consistent with less river-boundary/rate confounding in that formulation.

Table 4: Posterior pairwise correlations from the four full-data models. N/A denotes a fixed river DO, for which correlation with its constant value is undefined.

River boundary	Order	River DO vs rate	River DO vs LSR deficit	LSR deficit vs rate
Fixed	Zero	N/A	N/A	-0.181
Fixed	First	N/A	N/A	-0.397
Broad	Zero	0.838	0.404	0.175
Broad	First	0.544	0.281	-0.192

8 Completed sensitivity to LSR boundary and MRT uncertainty

Seven one-factor-at-a-time modifications of the selected *uncertain-river, first-order* model were fitted using the same 39 DO-likelihood wells. The reference case is recharge saturation at 15°C, an LSR oxygen survival prior $s \sim \mathrm{Beta}(1,1)$, and age log-SD $\sigma_t = 0.25$. The river DO prior and aquifer reaction law remain unchanged throughout. These are *conditional model sensitivity tests*; they are not source-specific transit-time models or further held-out predictive comparisons.

Table 5: Completed script-07 sensitivity analysis, from the user's reported output files. The percentages are relative to the *saved-fit* baseline median 0.0639 yr^{-1}. All eight fits passed the script-07 MCMC gates. Concentration medians are in mg L^{-1}.

Scenario	Median k_1 [yr^{-1}]	95% CrI	Change [%]	LSR WT DO	LSR deficit
Baseline: 15°C; Beta(1,1); MRT SD 0.25	0.0639	[0.0458,0.0857]	0.00	3.26	6.78
LSR saturation 10°C	0.0638	[0.0462,0.0854]	-0.120	3.25	8.02
LSR saturation 20°C	0.0639	[0.0459,0.0862]	+0.036	3.23	5.79
LSR survival Beta(2,2)	0.0638	[0.0458,0.0860]	-0.032	3.42	6.61
LSR survival Beta(2,5)	0.0632	[0.0451,0.0845]	-0.997	3.01	7.03
LSR survival Beta(5,2)	0.0660	[0.0479,0.0887]	+3.38	4.28	5.75
MRT log-SD 0.10	0.0610	[0.0448,0.0797]	-4.55	3.13	6.91
MRT log-SD 0.50	0.0757	[0.0520,0.108]	+18.5	3.90	6.13

Temperature changes from 10 to 20°C alter the median rate by less than 0.2%; alternative LSR survival priors alter it by approximately −1.0 to +3.4%. Water-table LSR DO remains near 3.2–3.3 mg L^{-1} across the temperature scenarios, whereas the surface-to-water-table oxygen deficit shifts

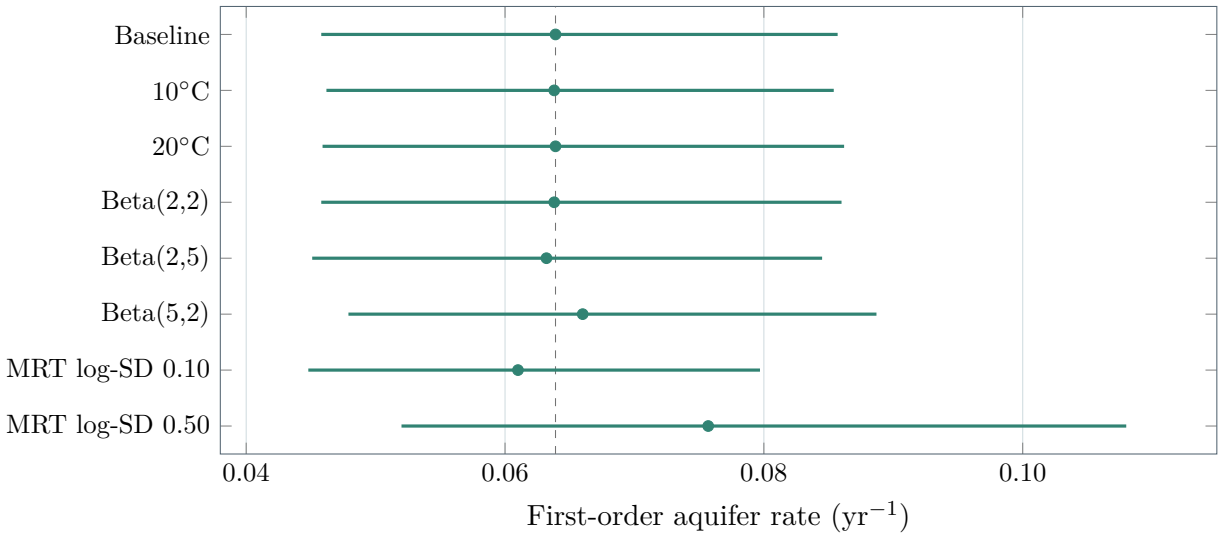


Figure 10: First-order rate posterior medians and 95% credible intervals for alternative assumptions. The dashed line marks the baseline median. These are *within-model* intervals, not intervals for scenario differences.

from 8.02 to 5.79 mg L^{-1}. This is partly a consequence of the *definition* of the deficit relative to an assumed saturation value. The prior Beta(5,2), favouring higher DO survival, shifts water-table DO to 4.28 mg L^{-1}, illustrating residual prior dependence.

Altering the *breadth of MRT uncertainty* has the largest effect on the median rate among these tests: reducing log-SD to 0.10 lowers k_1 by 4.55%, while increasing it to 0.50 raises k_1 by 18.5%. These changes do not demonstrate the effect of systematically longer or shorter groundwater ages. All eight baseline/sensitivity fits passed reported convergence checks (maximum $\widehat{R} = 1.01$, no divergent transitions or tree-depth hits, minimum E-BFMI 0.460 and minimum tail ESS 669). These checks establish numerical convergence, not hydrogeological validity.

Interpretation: the fitted first-order reaction rate is comparatively robust to the tested LSR saturation and survival-prior assumptions, but more sensitive to assumptions about uncertainty in groundwater ages. The tests do *not* address the more fundamental uncertainty in the source-specific residence times or progressive introduction of LSR along groundwater flowpaths.

9 Key hydrogeological limitation: progressive addition of LSR

9.1 What the mixing model captures

An increasing LSR contribution with river distance or along a downgradient groundwater flowpath is physically plausible in a system receiving diffuse land-surface recharge. The current mixing formulation, Equation (2), allows well-specific fractions f_i to vary from river-dominated values toward LSR-dominated values. Consequently, an increasing tracer-inferred LSR fraction along a sampled transect *can be reflected in the model*. However, it is not guaranteed: neither measured river distance nor a monotonic relationship with MRT is directly included.

Because DO is part of the joint likelihood, f_i is additionally informed by the reactive observations. It is therefore best described as a *jointly inferred source fraction constrained by isotope, chloride and DO data*, rather than as a wholly independent conservative tracer mixing estimate.

9.2 What the model omits

The present first-order DO equation (Equation (9)) implicitly assumes $t_{R,i} = t_{\mathrm{LSR},i} = t_i$. It calculates the age-reduced DO of *both* source endmembers over the same duration and mixes them only at the end. In a physically distributed recharge system, some LSR may have entered the aquifer close to the sampled well, while river-derived groundwater may have experienced a much longer journey. Assigning a full river-component MRT to young LSR would tend to give that LSR excess reaction exposure; whether the fitted k is biased upward or downward depends on which source the observed bulk MRT represents, their relative DO, and the fitted mixing fractions. A more flexible, still lumped representation would be

$$\widehat{\mathrm{DO}}_i = (1 - f_i) C_R \exp(-k t_{R,i}) + f_i C_{\mathrm{LSR},WT} \exp(-k t_{\mathrm{LSR},i}), \qquad t_{R,i} \neq t_{\mathrm{LSR},i} \text{ in general.} \tag{17}$$

For distributed recharge with an age spectrum, the more appropriate expression averages the exponential survival over a source-dependent travel-time distribution:

$$\widehat{\mathrm{DO}}_i = (1 - f_i) C_R \int_0^\infty e^{-kt}\, g_{R,i}(t)\, dt + f_i C_{\mathrm{LSR},WT} \int_0^\infty e^{-kt}\, g_{\mathrm{LSR},i}(t)\, dt. \tag{18}$$

Here $g_{R,i}$ and $g_{\mathrm{LSR},i}$ are source-conditioned groundwater age distributions, not quantities currently estimated in Stan. Equations (17) and (18) are conceptual extensions, **not fitted or validated results**. An observed bulk MRT cannot generally identify both component-specific travel-time distributions.

The impact on inferred rate and LSR boundary DO is not known a priori. Depending on relative source DO concentrations, recharge ages and mixing fractions, accounting for younger LSR may redistribute the inferred oxygen deficit between LSR entry conditions and subsequent aquifer reduction. It would be premature to assert the direction or magnitude of that bias without a sensitivity calculation.

9.3 Suggested verification before manuscript submission

1. Map or plot posterior f_i against *measured* distance from the river, depth and reported MRT, with credible intervals. Do not treat MRT itself as a direct surrogate for distance unless demonstrated. The current input script does not provide river-distance variables to Stan; these would be joined separately from well coordinates.
2. Use existing posterior median parameters for a *forward, no-refit* sensitivity calculation allowing $t_{\mathrm{LSR},i} < t_{R,i}$, under transparent scenarios. Report effects on predicted DO and whether the conclusions about k and boundary deficit are robust. This is a diagnostic, not a new calibrated result.
3. If sensitivity is substantial, seek independent constraints on source-specific ages or distributed recharge. Adding unconstrained age parameters to the present Bayesian model would risk losing the identifiability gained through parsimony.

10 Interpretation, strengths and limitations

10.1 What is supported

- The selected broad-river, first-order model estimates an effective aquifer DO depletion rate with a posterior interval that excludes zero in the stated formulation. The rate is stable under the tested river and LSR boundary conditions; its sensitivity to assumed MRT uncertainty is appreciably larger.

- The model requires substantially lower average LSR DO at aquifer entry than the assumed land-surface saturation concentration, conditional on the chosen two-source structure, residence-time model and shared LSR DO boundary.
- Grouped held-out validation favours first-order over zero-order kinetics under the fixed-river boundary, while the first-versus-zero advantage is less conclusive under the broad boundary. The broad river DO prior has a small CV RMSE penalty versus fixed DO, but is preferable as a physically honest boundary assumption.
- MCMC convergence is satisfactory for all four final full-data models; extending chains would not meaningfully address the dominant hydrogeological uncertainty.

10.2 What should not be overclaimed

- The inferred LSR DO deficit does not uniquely identify oxygen loss in the unsaturated zone. It is an effective difference between assumed rainfall DO and inferred water-table DO, conditional on the model.
- The fitted rate is an effective regional kinetic parameter. It need not be constant along individual aquifer flowpaths, across redox facies or between river-derived and LSR-derived waters.
- The model does not currently constrain source-specific ages or spatial recharge histories; increasing LSR fraction is not equivalent to progressive source addition in a reactive-transport calculation.
- DO predictions were held out *conditional on well tracer and MRT information*, not independently tested at new groundwater locations without those observations.
- Posterior means of losses across sampled wells are not flow-weighted or aquifer-wide oxygen mass budgets.

11 Potential manuscript wording

Methods paragraph (draft)

We implemented a joint Bayesian two-endmember mixing and dissolved-oxygen consumption model for Heretaunga groundwater. River recharge and land-surface recharge (LSR) fractions were estimated at individual wells using $\delta^{18}O$ and chloride, with DO observations contributing to the joint inference. River-entry DO was assigned a truncated Normal(7.5,1.5^2) $\mathrm{mg\,L^{-1}}$ prior in the selected first-order model; a fixed $7.5\,\mathrm{mg\,L^{-1}}$ boundary was retained for comparison. The DO of LSR reaching the water table was inferred relative to assumed freshwater saturation at the land surface (baseline 15°C). Both sources were exposed to the same well-specific effective MRT and one aquifer reaction law, with zero- and first-order kinetics fitted separately. DO at sites 2801 and 2130 was withheld but their conservative tracers retained, leaving 39 fitted DO observations after excluding site 16758. Four full-data models were compared using matched five-fold grouped cross-validation with held-out DO; a further seven full-data sensitivity fits varied LSR saturation temperature, LSR-survival priors and age uncertainty.

Results paragraph (draft)

The selected model used first-order aquifer oxygen consumption with uncertain river-entry DO. The subsequently reported saved-fit posterior median rate was $0.0639\,\mathrm{yr^{-1}}$ (95% credible interval 0.0458–0.0857); inferred LSR DO at the water table was $3.26\,\mathrm{mg\,L^{-1}}$ (1.31–5.57), corresponding to an LSR oxygen deficit of $6.78\,\mathrm{mg\,L^{-1}}$ (4.46–8.72). An earlier summary CSV of the same model

family reported a rate of $0.0632\,\mathrm{yr}^{-1}$ and should be reconciled with the saved posterior before publication. In grouped five-fold CV, the fixed-boundary first-order model marginally outperformed its uncertain-boundary counterpart by a median paired RMSE difference of $0.0342\,\mathrm{mg\,L}^{-1}$; this small predictive difference did not justify treating river-entry DO as exactly known. A seven-fit sensitivity analysis found first-order rate changes below 0.2% for 10–20°C LSR saturation, −1.0 to +3.4% for alternative LSR survival priors, and −4.6 to +18.5% for changes in MRT log-SD from 0.25 to 0.10 or 0.50. All sensitivity fits passed reported MCMC checks. Results remain conditional on common LSR boundary DO and simplified source age representation.

Discussion paragraph (draft)

The comparatively stable first-order rate under LSR boundary perturbations suggests that an effective aquifer oxygen-consumption parameter is estimable under the adopted tracer-constrained recharge model; the marked response to MRT uncertainty and unknown source-conditioned ages restricts physical interpretation. By contrast, the partition between LSR entry oxygen deficit and aquifer depletion may be more structurally sensitive. The mixing model permits increasing LSR contributions across groundwater wells but does not explicitly represent distributed recharge inputs or different residence times for river and LSR components. Because the model applies a common well-specific MRT to both source waters, the inferred LSR water-table DO is best interpreted as an effective boundary concentration, not a uniquely identified measure of oxygen consumption in the vadose zone. Future work should assess observed spatial trends in recharge fractions and the sensitivity of the fitted DO predictions to source-specific travel-time distributions.

12 Reproducibility and figure use

The final scripts are maintained in Oxygen-Reduction-Joint-Model. Following the Positron renumbering, the active workflow is:

1. `scripts/00_setup.R`: data validation and shared Stan inputs (no sampling).
2. `scripts/02_river_DO_prior_sensitivity.R`: the four full-data posterior fits (two river boundaries by two kinetic orders).
3. `scripts/03_river_DO_grouped_CV.R`: grouped held-out DO validation, with resumable fit cache.
4. `scripts/01_DO_plotting.R`: summary and cross-validation plots, based on scripts 02 and 03 outputs.
5. `scripts/04_DO_posterior_distributions.R`: one preferred first-order rate posterior density and posterior summary from the saved draws.
6. `scripts/05_DO_kinetic_trajectoriesandsummary.R`: five numbered DO–MRT figures (mixing-adjusted first-order survival; observed versus fitted total DO; raw DO versus river-only reference; inferred river-attributable DO; and additive river/LSR components), plus a four-parameter posterior summary. All five appear in the report. Script 05 reads saved CmdStanR fits, writes PNGs under `plots/final_39well/`, and does not run Stan.
7. `scripts/06_DO_LSR_MRT_sensitivity.R`: seven LSR/MRT sensitivity fits, with cache protection.

The Stan model remains `stan/04_river_DO_prior_sensitivity.stan`. These were file and plotting-workflow changes, not changes to the fitted scientific model or previously validated posterior draws. Output directories intentionally retain historical names such as `outputs/04_river_DO_prior/` and `outputs/05_river_DO_grouped_CV/`, even though the runner scripts have

been renumbered. Retired scripts are preserved in `archive/`. The root `scripts/tests.R` is an optional plotting scratch script, not a required analysis step.

The saved fitted RDS objects, posterior CSVs, generated PNGs and private measurements reside only in the user's local working copy and are gitignored. **All five DO–MRT figures are incorporated through script 05's output paths, not phone-uploaded images.** Figures 1 and 2 are conceptual; Figures 3 and 10 are based on reported tables; Figure 4 is an illustrative application of median parameters. The plotting script reads original well-level posterior draws saved locally, but those objects are not packaged with this report.

Compiling the document

From the repository root, regenerate all five figures in Positron using:

```
source("scripts/05_DO_kinetic_trajectories and summary.R")
```

This reads the existing preferred CmdStanR RDS and produces exact output files under `plots/final_39well/`; no refitting is required. Put this `.tex` file either in the repository root or in `docs/`, and compile from the directory containing it with `pdflatex` twice (or `latexmk -pdf`). Its file-resolution logic accepts either `plots/final_39well/` or `../plots/final_39well/`. If the PNG files are missing, the compiled PDF shows an explicit missing-output placeholder rather than silently inserting old uploaded images. Recompile the PDF whenever the script outputs change: PDF images do not update automatically.

A Detailed model settings and data constraints

Table 6: Selected implementation settings, sourced from the final model code.

Setting	Value or interpretation
Conservative source tracers	δ^{18}O and chloride only; no Na, excess air or N_2 in fitted likelihood
Source fractions	One $f_i \in [0,1]$ per well, uniform Beta(1,1) prior
LSR survival	One global fraction $s \in [0,1]$, Beta(1,1) prior
River DO primary / comparator	Truncated Normal(7.5, 1.5^2) / fixed 7.5 mg L^{-1}
Initial LSR DO	Approx. saturated freshwater DO at 15°C, assumed atmospheric pressure ratio 1
Age uncertainty	Lognormal with log-SD 0.25 at baseline (tested 0.10 and 0.50), adjusted to preserve arithmetic reported MRT
Reaction rates	Zero-order k_0 or first-order k_1, fitted in separate runs
DO observation likelihood	Student-t, four degrees of freedom; scale $\sqrt{0.2^2 + \sigma_{DO}^2}$ mg L^{-1}
Full-data sampling	Four chains, 1,400 warm-up plus 1,400 retained iterations per chain
Cross-validation	Five matched grouped folds; 39 scored groundwater DO observations per variant
Bootstrap	2,000 resamples of matched well groups for paired RMSE and MAE differences
Additional sensitivity fits	Seven independent four-chain fits, varying recharge temperature, survival prior, and MRT log-SD

B Script-generated DO–MRT figures and exact filenames

Active script 05 writes five consecutively numbered PNG outputs into `plots/final_39well/`. All five are included in the main report:

1. `01_DO_mixing_adjusted_decay_2801_masked_no2130.png` (Figure 5);
2. `02_DO_observed_vs_fitted_mixed_2801_masked_no2130.png` (Figure 6);
3. `03_DO_river_reference_raw_samples_2801_masked_no2130.png` (Figure 7);
4. `04_DO_river_contribution_2801_masked_no2130.png` (Figure 8);
5. `05_DO_fitted_components_2801_masked_no2130.png` (Figure 9).

The four-parameter posterior summary is written separately as `06_DO_primary_posterior_summary_2801_masked_no2130.md`. All figure files and saved model fits are kept locally (git-ignored), and the report uses no uploaded screenshot substitutes. Plot 1's normalised observed values are conditional on the joint mixing fit, so its credible whiskers are not an independent measurement of kinetic survival. Plot 2 uses posterior predictive whiskers; Plot 4 displays inferred fractional river contribution relative to an unweighted pure-river reference; and Plot 5 uses additive posterior mean contributions with credible intervals for fitted mean DO. The conceptual two-source decay graphic remains separate as Figure 4.

C Provenance and editorial checklist

The numerical posterior estimates, model settings, correlations, paired CV comparisons and seven-fit sensitivity results are drawn from user-reported November 2026 outputs and reviewed repository code. Stan was not rerun for this report. The earlier 0.0632 versus saved-fit 0.0639 yr^{-1} rate-summary discrepancy remains flagged pending reconciliation. Script 05 now generates five consecutively numbered figures and this report includes all five. The plotting changes do not alter Stan, the fitted likelihood, posterior fits or grouped CV. The report source uses the exact script-generated PNG paths and must be recompiled after local plot generation.

Before manuscript submission, regenerate all five figure outputs locally, check the five panels and their captions, reconcile the posterior summaries between RDS and the older CSV, supply relevant maps and absolute CV scores, and add literature references. Source contributions and mixing-adjusted observations remain model-conditioned rather than independent validation of the shared effective MRT assumption.